

## Experimental approaches for the treatment of malignant gliomas.

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Malignant gliomas, which include glioblastomas and anaplastic astrocytomas, are the most common primary tumors of the brain. Over the past 30 years, the standard treatment for these tumors has evolved to include maximal safe surgical resection, radiation therapy and temozolomide chemotherapy. While the median survival of patients with glioblastomas has improved from 6 months to 14.6 months, these tumors continue to be lethal for the vast majority of patients. There has, however, been recent substantial progress in our mechanistic understanding of tumor development and growth. The translation of these genetic, epigenetic and biochemical findings into therapies that have been tested in clinical trials is the subject of this review.

## Assessment and prediction of glioblastoma therapy response: challenges and opportunities.

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Glioblastoma is the most aggressive type of primary adult brain tumour. The median survival of patients with glioblastoma remains approximately 15 months, and the 5-year survival rate is <10%. Current treatment options are limited, and the standard of care has remained relatively constant since 2011. Over the last decade, a range of different treatment regimens have been investigated with very limited success. Tumour recurrence is almost inevitable with the current treatment strategies, as glioblastoma tumours are highly heterogeneous and invasive. Additionally, another challenging issue facing patients with glioblastoma is how to distinguish between tumour progression and treatment effects, especially when relying on routine diagnostic imaging techniques in the clinic. The specificity of routine imaging for identifying tumour progression early or in a timely manner is poor due to the appearance similarity of post-treatment effects. Here, we concisely describe the current status and challenges in the assessment and early prediction of therapy response and the early detection of tumour progression or recurrence. We also summarize and discuss studies of advanced approaches such as quantitative imaging, liquid biomarker discovery and machine intelligence that hold exceptional potential to aid in the therapy monitoring of this malignancy and early prediction of therapy response, which may decisively transform the conventional detection methods in the era of precision medicine.

## A Prognostic Model for Glioblastoma Patients Treated With Standard Therapy Based on a Prospective Cohort of Consecutive Non-Selected Patients From a Single Institution.

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Glioblastoma patients administered standard therapies, comprising maximal surgical resection, radiation therapy with concomitant and adjuvant temozolomide, have a variable prognosis with a median overall survival of 15-16 months and a 2-year overall survival of 30%. The aim of this study was to develop a prognostic nomogram for overall survival for glioblastoma patients treated with standard therapy outside clinical trials. The study included 680 consecutive, non-selected glioblastoma patients administered standard therapy as primary treatment between the years 2005 and 2016 at Rigshospitalet, Copenhagen, Denmark. The prognostic model was generated employing multivariate Cox regression

analysis modeling overall survival. The following poor prognostic factors were included in the final prognostic model for overall survival: Age (10-year increase: HR = 1.18, 95% CI: 1.08-1.28,  $p < 0.001$ ), ECOG performance status (PS) 1 vs. 0 (HR = 1.30, 95% CI: 1.07-1.57,  $p = 0.007$ ), PS 2 vs. 0 (HR = 2.99, 95% CI: 1.99-4.50,  $p < 0.001$ ), corticosteroid use (HR = 1.42, 95% CI: 1.18-1.70,  $p < 0.001$ ), multifocal disease (HR = 1.63, 95% CI: 1.25-2.13,  $p < 0.001$ ), biopsy vs. resection (HR = 1.35, 95% CI: 1.04-1.72,  $p = 0.02$ ), un-methylated promoter of the MGMT (O6-methylguanine-DNA methyltransferase) gene (HR = 1.71, 95% CI: 1.42-2.04,  $p < 0.001$ ). The model was validated internally and had a concordance index of 0.65. A nomogram for overall survival was established. This model can be used for risk stratification and treatment planning, as well as improve enrollment criteria for clinical trials.

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## Diosmin Inhibits Glioblastoma Growth through Inhibition of Autophagic Flux.

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Diosmin, a natural flavone glycoside acquired through dehydrogenation of the analogous flavanone glycoside hesperidin, is plentiful in many citrus fruits. Glioblastoma multiforme (GBM) is the most malignant primary brain tumor; the average survival time of GBM patients is less than 18 months after standard treatment. The present study demonstrated that diosmin, which is able to cross the blood-brain barrier, inhibited GBM cell growth in vitro and in vivo. Diosmin also impeded migration and invasion by GBM8401 and LN229 GBM cells by suppressing epithelial-mesenchymal transition, as indicated by increased expression of E-cadherin and decreased expression of Snail and Twist. Diosmin also suppressed autophagic flux, as indicated by increased expression of LC3-II and p62, and induced cell cycle arrest at G1 phase. Importantly, diosmin did not exert serious cytotoxic effects toward control SVG-p12 astrocytes, though it did reduce astrocyte viability at high concentrations. These findings provide potentially helpful support to the development of new therapies for the treatment of GBM.

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## MicroRNAs as Multifaceted Players in Glioblastoma Multiforme.

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Glioblastoma multiforme (GBM) is the most common and inevitably lethal primary brain tumor, with a median survival rate of only 15 months from diagnosis. The current standard treatment involves maximal surgical resection flanked by radiotherapy and chemotherapy with the alkylating agent temozolomide. However, even such aggressive treatment is never curative, and recurrent tumors always arise, commonly in more aggressive, chemo- and radio-resistant forms, leading to untreatable and deadly tumors. MicroRNAs, recognized major players in cancer, are deeply involved in GBM, as shown by more than a decade of studies. In this review, we revise the main milestones of MicroRNA studies in GBM, and the latest relevant discoveries in this field. Examples are given of MicroRNAs working as "oncomiRs" or tumor suppressors, with specific connections with GBM clinical subtypes, patients' survival, and resistance to therapies. As the interaction of GBM cells with the microenvironment was proven as a key determinant of tumor growth, the role of MicroRNAs in GBM microenvironment, tumor angiogenesis, and tumor-secreted microvesicles is also reviewed. Finally, we discuss the latest findings presenting MicroRNAs as possible therapeutic targets for GBM, or their use as circulating biomarkers in diagnosis and prognosis.

## Recent advances in glioblastoma multiforme therapy: A focus on autophagy regulation.

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Despite conventional treatment options including chemoradiation, patients with the most aggressive primary brain tumor, glioblastoma multiforme (GBM), experience an average survival time of less than 15 months. Regarding the malignant nature of GBM, extensive research and discovery of novel treatments are urgently required to improve the patients' prognosis. Autophagy, a crucial physiological pathway for the degradation and recycling of cell components, is one of the exciting targets of GBM studies. Interventions aimed at autophagy activation or inhibition have been explored as potential GBM therapeutics. This review, which delves into therapeutic techniques to block or activate autophagy in preclinical and clinical research, aims to expand our understanding of available therapies battling GBM.

## Dendritic cell vaccination for glioblastoma multiforme: review with focus on predictive factors for treatment response.

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Glioblastoma multiforme (GBM) is the most common and most aggressive type of primary brain cancer. Since median overall survival with multimodal standard therapy is only 15 months, there is a clear need for additional effective and long-lasting treatments. Dendritic cell (DC) vaccination is an experimental immunotherapy being tested in several Phase I and Phase II clinical trials. In these trials, safety and feasibility have been proven, and promising clinical results have been reported. On the other hand, it is becoming clear that not every GBM patient will benefit from this highly personalized treatment. Defining the subgroup of patients likely to respond to DC vaccination will position this option correctly amongst other new GBM treatment modalities, and pave the way to incorporation in standard therapy. This review provides an overview of GBM treatment options and focuses on the currently known prognostic and predictive factors for response to DC vaccination. In this way, it will provide the clinician with the theoretical background to refer patients who might benefit from this treatment.

## Innovative Approaches to Brain Cancer: The Use of Magnetic Resonance-guided Focused Ultrasound in Glioma Therapy.

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Gliomas are a wide group of common brain tumors, with the most aggressive type being glioblastoma multiforme (GBM), with a 5-year survival rate of less than 5% and a median survival time of approximately 12-14 months. The standard treatment of GBM includes surgical excision, radiotherapy, and chemotherapy with temozolomide (TMZ). However, tumor recurrence and progression are common. Therefore, more effective treatment for GBM should be found. One of the main obstacles to the treatment of GBM and other gliomas is the blood-brain barrier (BBB), which impedes the penetration of antitumor chemotherapeutic agents into glioblastoma cells. Nowadays, one of the most promising novel methods for glioma treatment is Magnetic Resonance-guided Focused Ultrasound (MRgFUS). Low-intensity FUS causes the BBB to open transiently, which allows better drug delivery to the brain tissue. Under magnetic resonance guidance, ultrasound waves can be precisely directed to the tumor area to prevent side effects in healthy tissues. Through the open BBB, we can deliver targeted

chemotherapeutics, anti-tumor agents, immunotherapy, and gene therapy directly to gliomas. Other strategies for MRgFUS include radiosensitization, sonodynamic therapy, histotripsy, and thermal ablation. FUS can also be used to monitor the treatment and progression of gliomas using blood-based liquid biopsy. All these methods are still under preclinical or clinical trials and are described in this review to summarize current knowledge and ongoing trials.

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## Cellular-based immunotherapies for patients with glioblastoma multiforme.

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Treatment of patients with glioblastoma multiforme (GBM) remains to be a challenge with a median survival of 14.6 months following diagnosis. Standard treatment options include surgery, radiation therapy, and systemic chemotherapy with temozolomide. Despite the fact that the brain constitutes an immunoprivileged site, recent observations after immunotherapies with lysate from autologous tumor cells pulsed on dendritic cells (DCs), peptides, protein, messenger RNA, and cytokines suggest an immunological and even clinical response from immunotherapies. Given this plethora of immunomodulatory therapies, this paper gives a structure overview of the state-of-the art in the field. Particular emphasis was also put on immunogenic antigens as potential targets for a more specific stimulation of the immune system against GBM.

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## In vitro anticancer drug test: A new method emerges from the model of glioma stem cells.

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Glioblastoma multiforme (GBM) is a grade IV astrocytoma and the most common malignant brain tumor. Current therapies provide a median survival of 12-15 months after diagnosis, due to the high recurrence rate. The failure of current therapies may be due to the presence, within the tumor, of cells characterized by enhanced self-renewal capacity, multilineage differentiation potential and elevated invasive behavior, called glioma stem cells (GSCs). To evaluate the pharmacological efficacy of selected drugs on six GSC lines, we set up a multiple drug responsivity assay based on the combined evaluation of cytomorphological and functional parameters, including the analysis of polymorphic nuclei, mitotic index and cell viability. In order to understand the real pharmacological efficacy of the tested drugs, we assigned a specific drug responsivity score to each GSC line, integrating the data produced by multiple assays. In this work we explored the antineoplastic effects of paclitaxel (PTX), an inhibitor of microtubule depolymerization, utilized as standard treatment in several cancers, and of valproic acid (VPA), an inhibitor of histone deacetylases (HDACs) with multiple anticancer properties. We classified the six GSC lines as responsive or resistant to these drugs, on the basis of their responsivity scores. This method can also be useful to identify the best way to combine two or more drugs. In particular, we utilized the pro-differentiating effect of VPA to improve the PTX effectiveness and we observed a significant reduction of cell viability compared to single treatments.